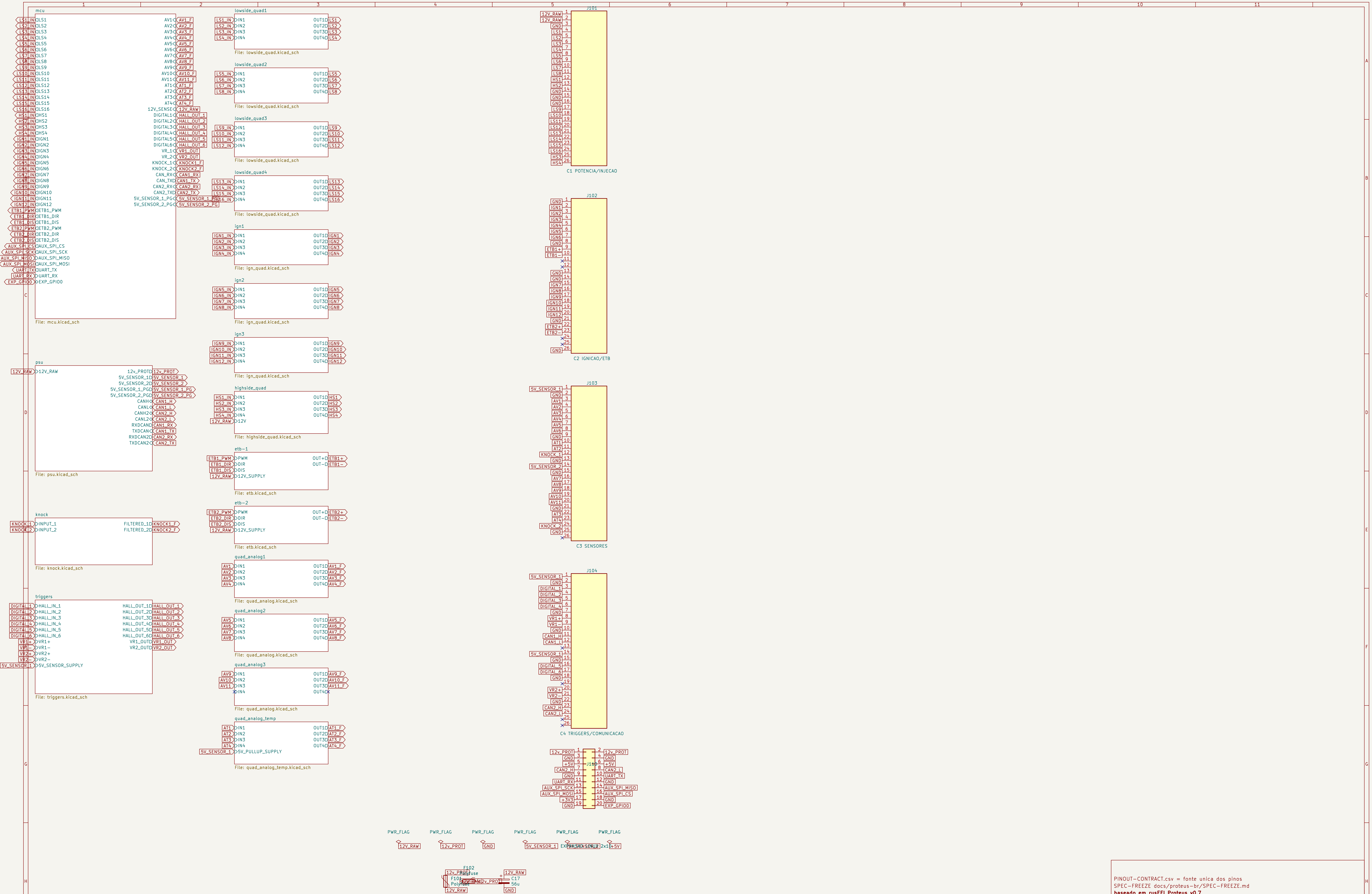


PROTEUS-BR v1 — raiz. Conectores TE Superseal 1.0 26v (headers 6437288-6). Keying: C1=key1, C2=key2, C3=key3, C4=key1 repetido (decisao 5 — afastado do C1, housing de cor distinta).

Limite assumido: 5 A continuos por pino (PREMISSA conservadora; terminal 18 AWG 3-1447221-3). Potencia sempre em pinos paralelados (12V_RAW e GND no C1).

Populacao fase 1 por instancia (DNP gerenciado na BOM de montagem — KiCad nao tem DNP por instancia de folha): lowside_quad3/4 (LS9-16) NAO populadas; ign2/ign3 (IGN5-12) NAO populadas; quad_analog3 (AV9-11) NAO populada; highside_quad e etb-1/etb-2 inteiras NAO populadas.

ETB: 1 pino por fase no C2 (decisao 8) — pico 6 A transitorio aceito; reservas 11/12/24/25 do C2 seladas com tampao 4-1437284-3.

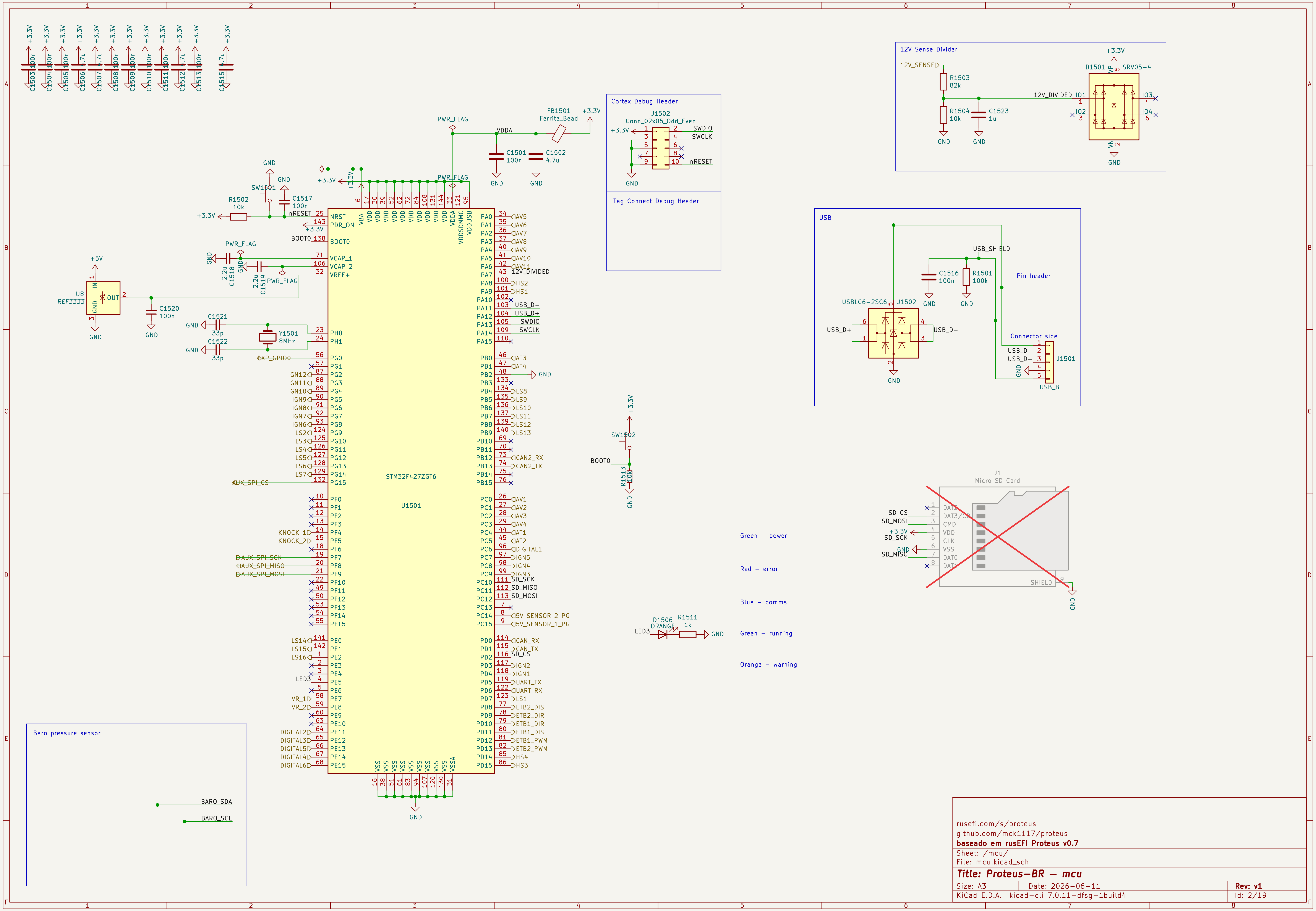
12V: C1.1-2 (12V_RAW, fusivel externo no chicote) -> TVS SM15T33CA (psu) + bulk C17; polyfuses F101||F102 -> 12v_PROT (buck, trackers, expansao). Highside/ETB/roda-livre direto no 12V_RAW.

PINOUT-CONTRACT.csv = fonte unica dos pinos
SPEC-FREEZE docs/proteus-br/SPEC-FREEZE.md
baseado em rusEFI Proteus v0.7

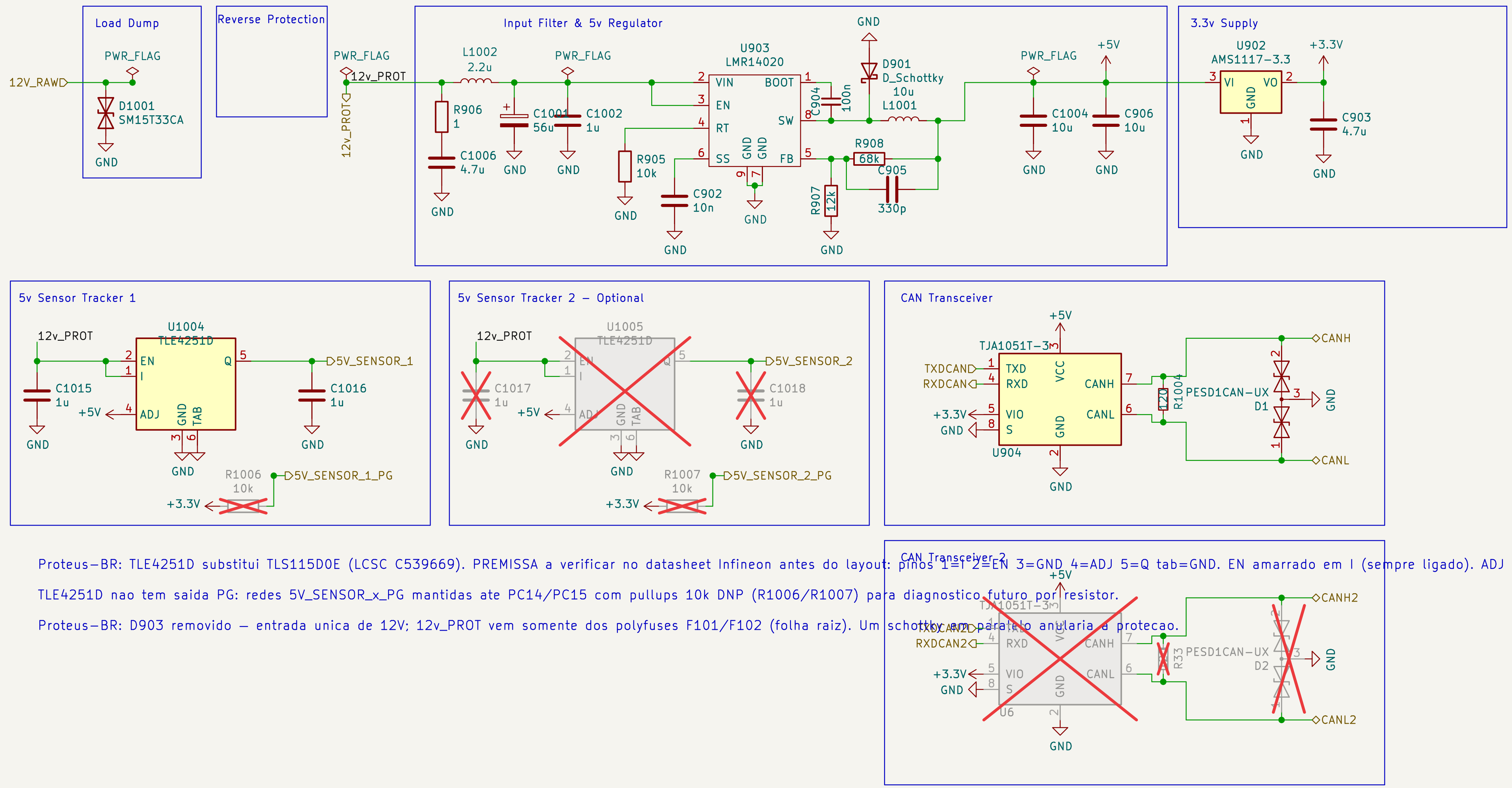
Sheet: /
File: proteus-br.kicad_sch

Title: Proteus-BR

Size: A2	Date: 2026-06-11	**Rev: v1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/19



rusefi.com/s/proteus
github.com/mck1117/proteus
baseado em rusEFI Proteus v0.7
Sheet: /mcu/
File: mcu.kicad_sch
Title: Proteus-BR – mcu
Size: A3 | Date: 2026-06-11 | Rev: v1
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 2/19



Proteus-BR: TLE4251D substitui TLS115D0E (LCSC C539669). PREMISSA a verificar no datasheet Infineon antes do layout: pinos 1=I 2=EN 3=GND 4=ADJ 5=Q tab=GND. EN amarrado em I (sempre ligado). ADJ segue o 5V do buck (tracki

TLE4251D nao tem saida PG: redes 5V_SENSOR_x_PG mantidas ate PC14/PC15 com pullups 10k DNP (R1006/R1007) para diagnostico futuro por resistor.

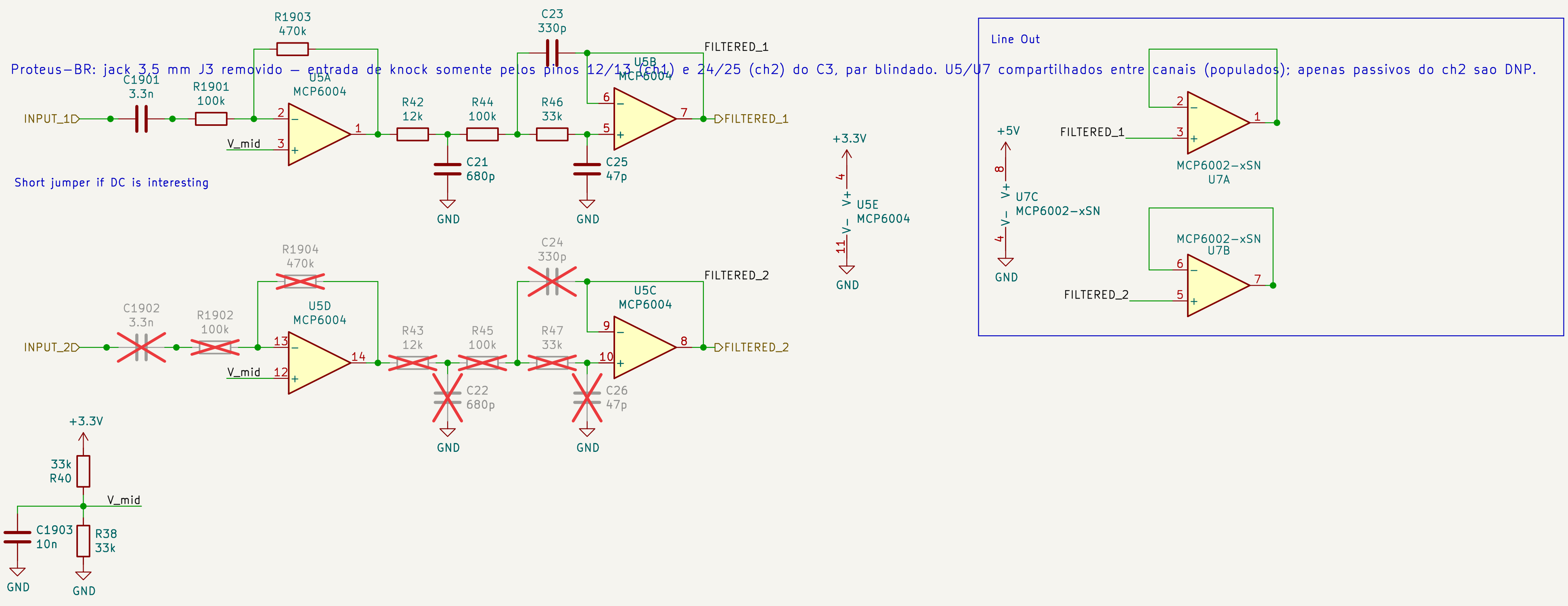
Proteus-BR: D903 removido — entrada unica de 12V; 12v_PROT vem somente dos polyfuses F101/F102 (folha raiz). Um schottky em paralelo anularia a protecao.

rusefi.com/s/proteus
github.com/mck1117/proteus
baseado em rusEFI Proteus v0.7

Sheet: /psu/
File: psu.kicad_sch

Title: Proteus-BR — psu

Size: A4	Date: 2026-06-11	**Rev: v1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/19

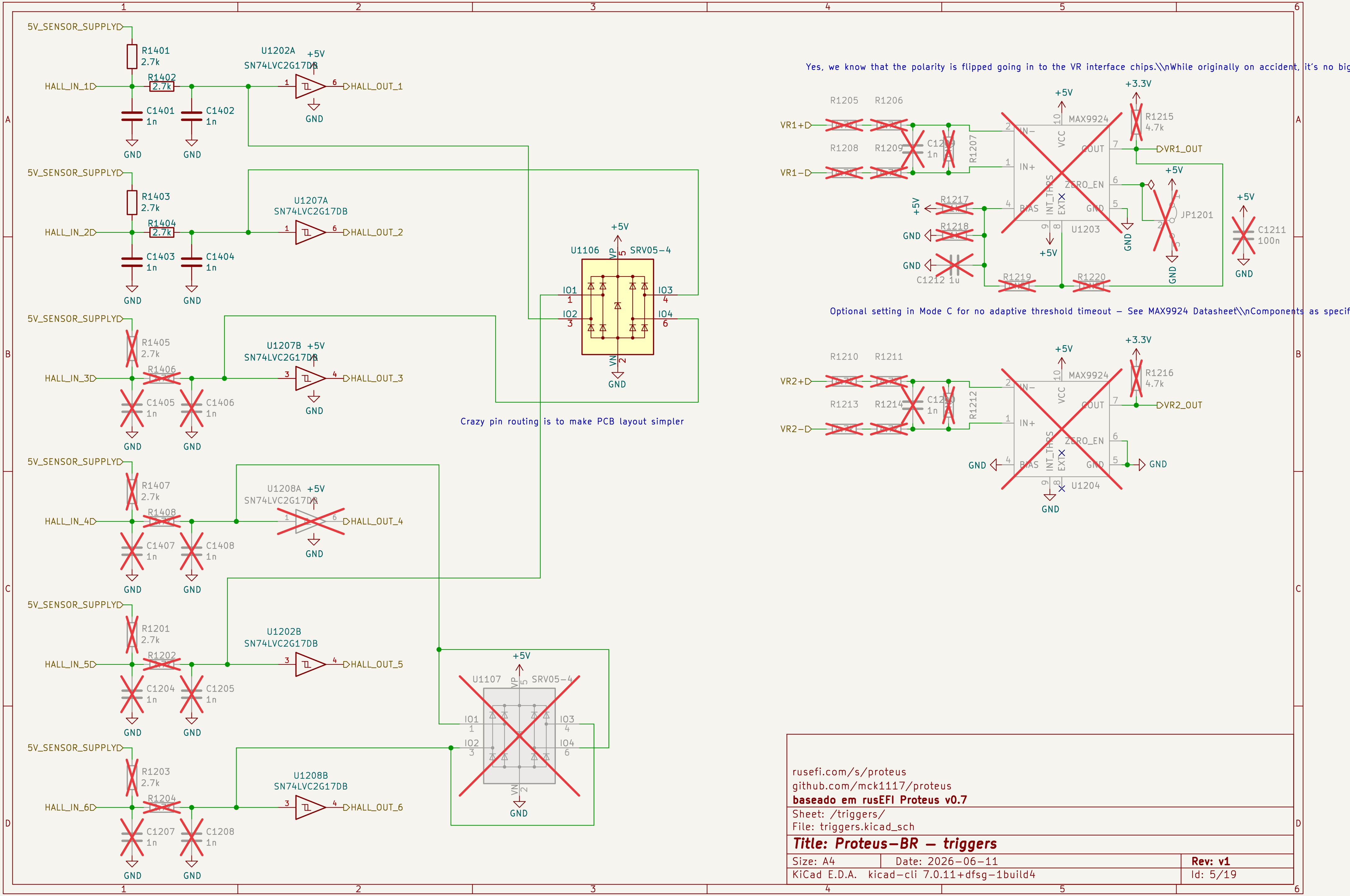


rusefi.com/s/proteus
github.com/mck1117/proteus
baseado em rusEFI Proteus v0.7

Sheet: /knock/
File: knock.kicad_sch

Title: Proteus-BR — knock

Size: A4	Date: 2026-06-11	**Rev: v1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/19

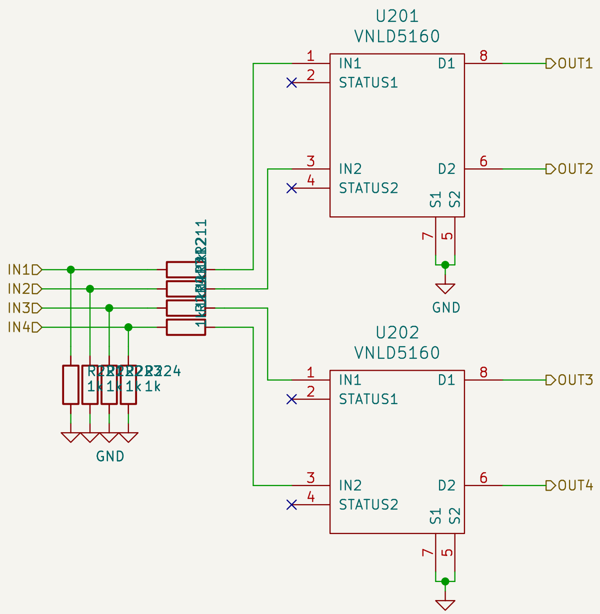


rusefi.com/s/proteus
github.com/mck1117/proteus
baseado em rusEFI Proteus v0.7

Sheet: /triggers/
File: triggers.kicad_sch

Title: Proteus-BR — triggers

Size: A4 Date: 2026-06-11 **Rev: v1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 Id: 5/19



Proteus-BR: arrays RN 1k convertidos em discretos 0603 (retrabalho manual mais facil). Serie 1k + pulldown 1k por canal, iguais ao original.

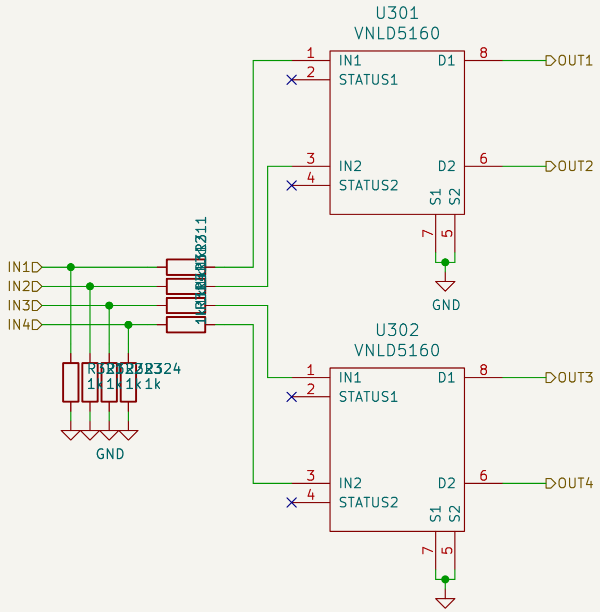
Locate output caps near connector pin

rusefi.com/s/proteus
github.com/mck1117/proteus
baseado em rusEFI Proteus v0.7

Sheet: /lowside_quad1/
File: lowside_quad.kicad_sch

Title: Proteus-BR — lowside_quad

Size: A4 Date: 2026-06-11 **Rev: v1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 Id: 6/19



Proteus-BR: arrays RN 1k convertidos em discretos 0603 (retrabalho manual mais facil). Serie 1k + pulldown 1k por canal, iguais ao original.

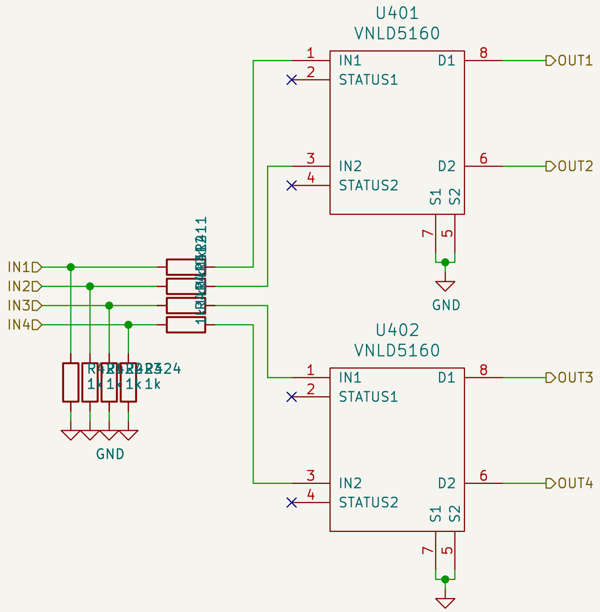
Locate output caps near connector pin

rusefi.com/s/proteus
github.com/mck1117/proteus
baseado em rusEFI Proteus v0.7

Sheet: /lowside_quad2/
File: lowside_quad.kicad_sch

Title: Proteus-BR — lowside_quad

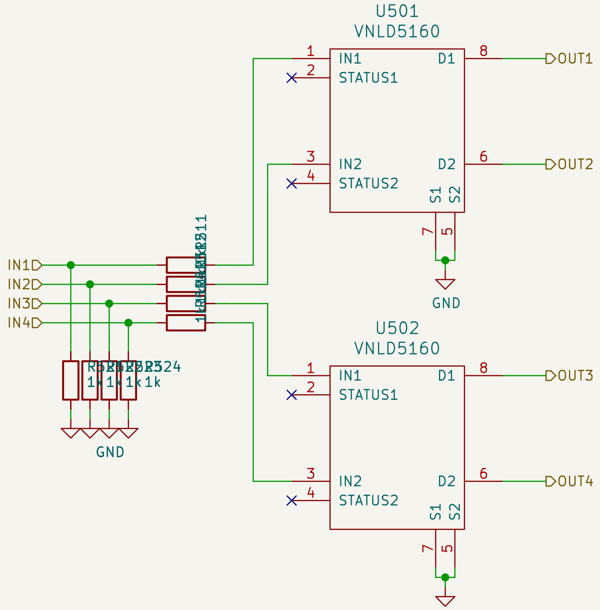
Size: A4	Date: 2026-06-11	**Rev: v1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 7/19



Proteus-BR: arrays RN 1k convertidos em discretos 0603 (retrabalho manual mais facil). Serie 1k + pulldown 1k por canal, iguais ao original.

Locate output caps near connector pin

rusefi.com/s/proteus
github.com/mck1117/proteus
baseado em rusEFI Proteus v0.7
Sheet: /lowside_quad3/
File: lowside_quad.kicad_sch
Title: Proteus-BR — lowside_quad
Size: A4 | Date: 2026-06-11 | **Rev: v1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 8/19



Proteus-BR: arrays RN 1k convertidos em discretos 0603 (retrabalho manual mais facil). Serie 1k + pulldown 1k por canal, iguais ao original.

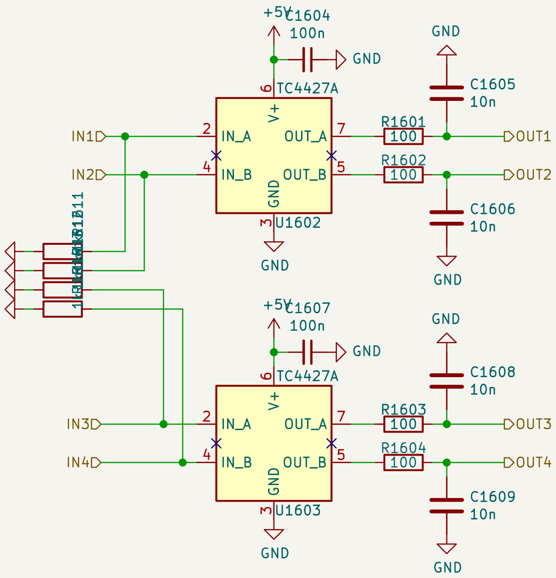
Locate output caps near connector pin

rusefi.com/s/proteus
github.com/mck1117/proteus
baseado em rusEFI Proteus v0.7

Sheet: /lowside_quad4/
File: lowside_quad.kicad_sch

Title: Proteus-BR — lowside_quad

Size: A4	Date: 2026-06-11	**Rev: v1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 9/19



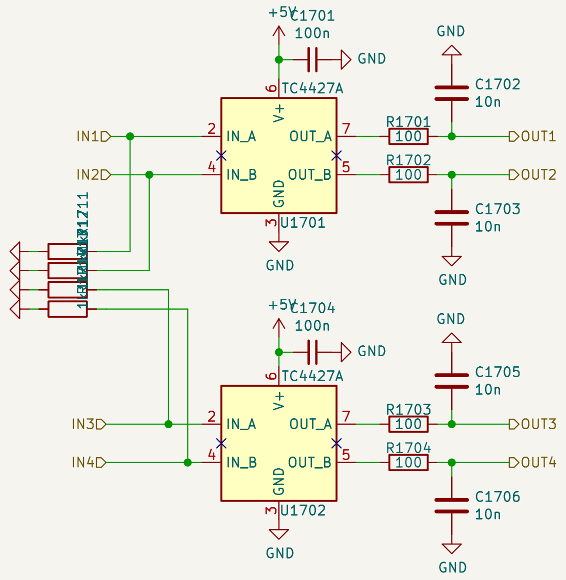
Proteus-BR: TC4427ACOA713 (LCSC C144234) substitui MIC4427 (indisponivel na LCSC) — pin-compativel, mesma familia. Serie 100R em 0805.

rusefi.com/s/proteus
github.com/mck1117/proteus
baseado em rusEFI Proteus v0.7

Sheet: /ign1/
File: ign_quad.kicad_sch

Title: Proteus-BR — ign_quad

Size: A4	Date: 2026-06-11	**Rev: v1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 10/19

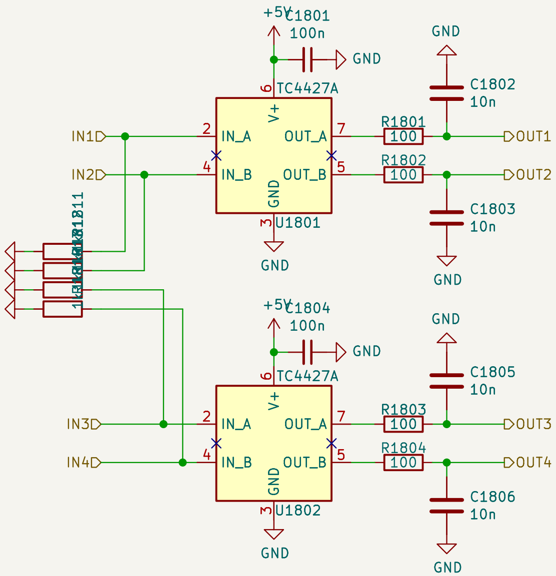


Proteus-BR: TC4427ACOA713 (LCSC C144234) substitui MIC4427 (indisponivel na LCSC) — pin-compativel, mesma familia. Serie 100R em 0805.

rusefi.com/s/proteus
github.com/mck1117/proteus
baseado em rusEFI Proteus v0.7
Sheet: /ign2/
File: ign_quad.kicad_sch

Title: Proteus-BR — ign_quad

Size: A4	Date: 2026-06-11	**Rev: v1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 11/19



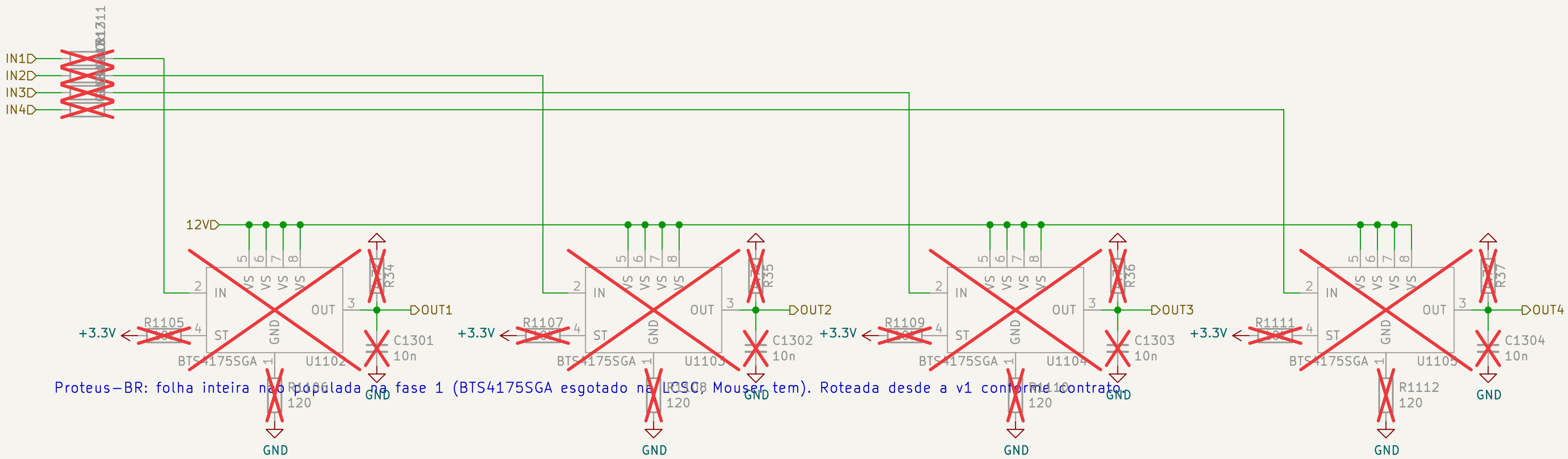
Proteus-BR: TC4427ACOA713 (LCSC C144234) substitui MIC4427 (indisponivel na LCSC) — pin-compativel, mesma familia. Serie 100R em 0805.

rusefi.com/s/proteus
github.com/mck1117/proteus
baseado em rusEFI Proteus v0.7

Sheet: /ign3/
File: ign_quad.kicad_sch

Title: Proteus-BR — ign_quad

Size: A4	Date: 2026-06-11	**Rev: v1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 12/19



Proteus-BR: folha inteira não populada na fase 1 (BTS4175SGA esgotado na LCSC; Mouser tem). Roteada desde a v1 conforme contrato.

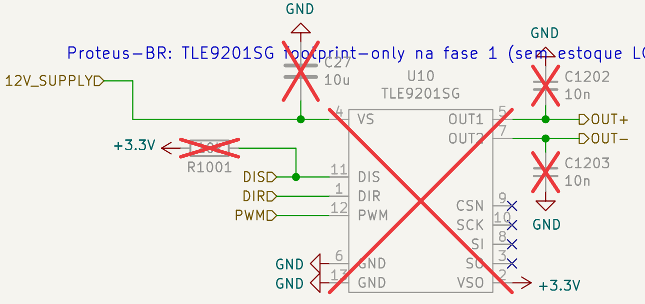
rusefi.com/s/proteus
github.com/mck1117/proteus
baseado em rusEFI Proteus v0.7

Sheet: /highside_quad/
File: highside_quad.kicad_sch

Title: Proteus-BR — highside_quad

Size: A4	Date: 2026-06-11	Rev: v1
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 13/19

Proteus-BR: TLE9201SG footprint-only na fase 1 (sem estoque LCSC; Mouser TLE9201SGAUMA1, gemeo industrial IFX9201SG LCSC C112633).

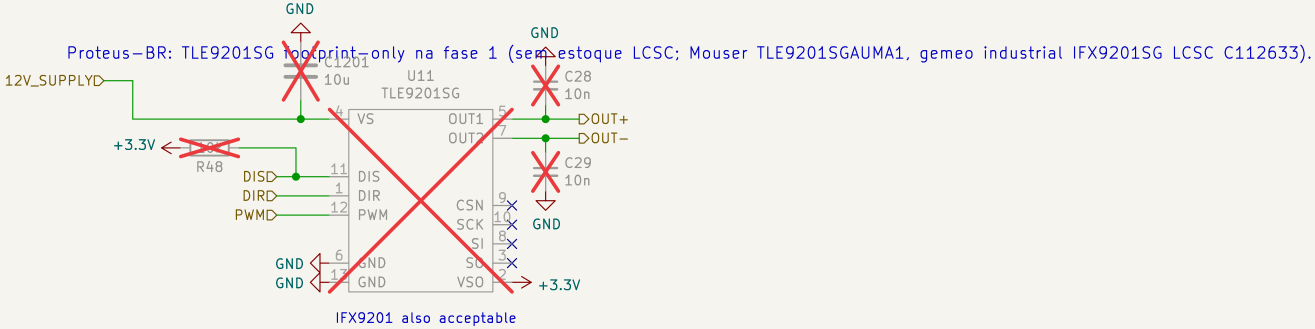


IFX9201 also acceptable

rusefi.com/s/proteus
github.com/mck1117/proteus
baseado em rusEFI Proteus v0.7
Sheet: /etb-1/
File: etb.kicad_sch

Title: Proteus-BR — etb

Size: A4	Date: 2026-06-11	**Rev: v1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 14/19

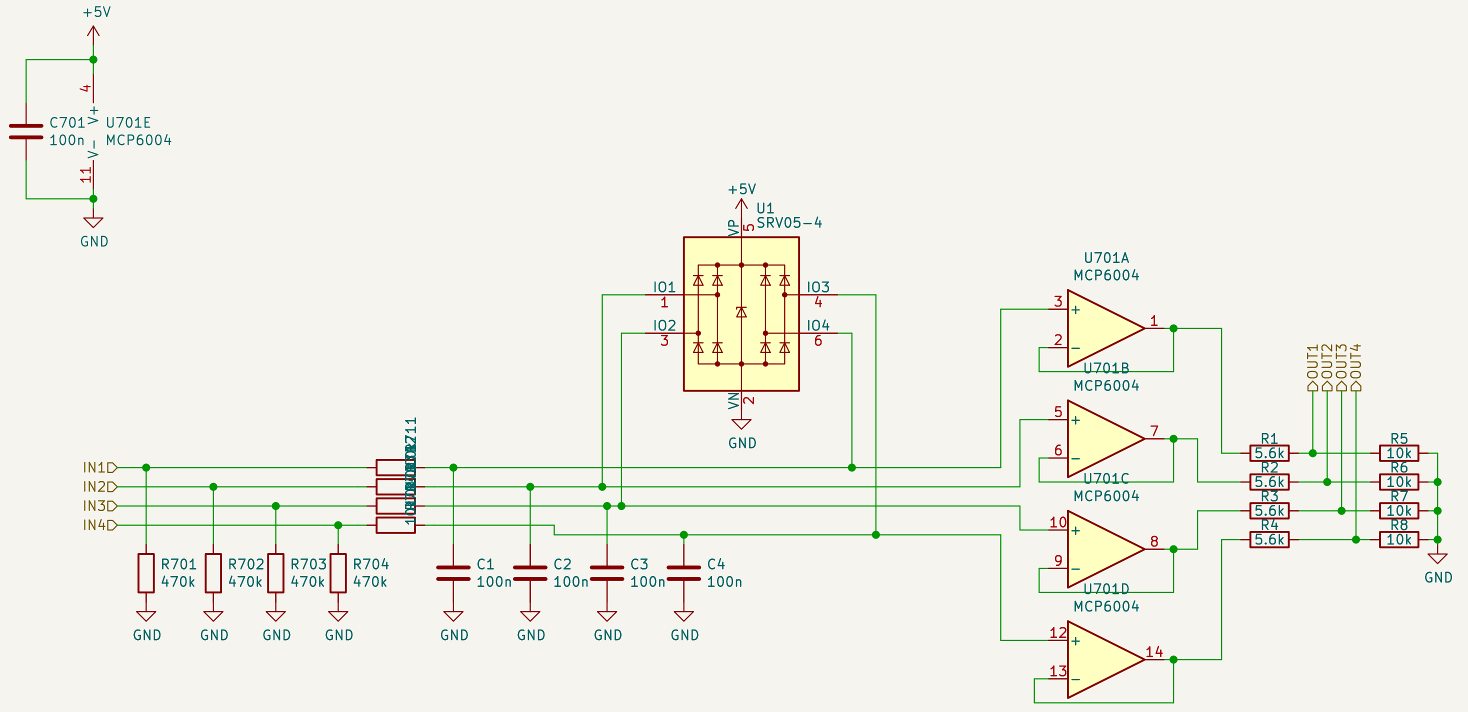


rusefi.com/s/proteus
github.com/mck1117/proteus
baseado em rusEFI Proteus v0.7

Sheet: /etb-2/
File: etb.kicad_sch

Title: Proteus-BR — etb

Size: A4	Date: 2026-06-11	**Rev: v1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 15/19



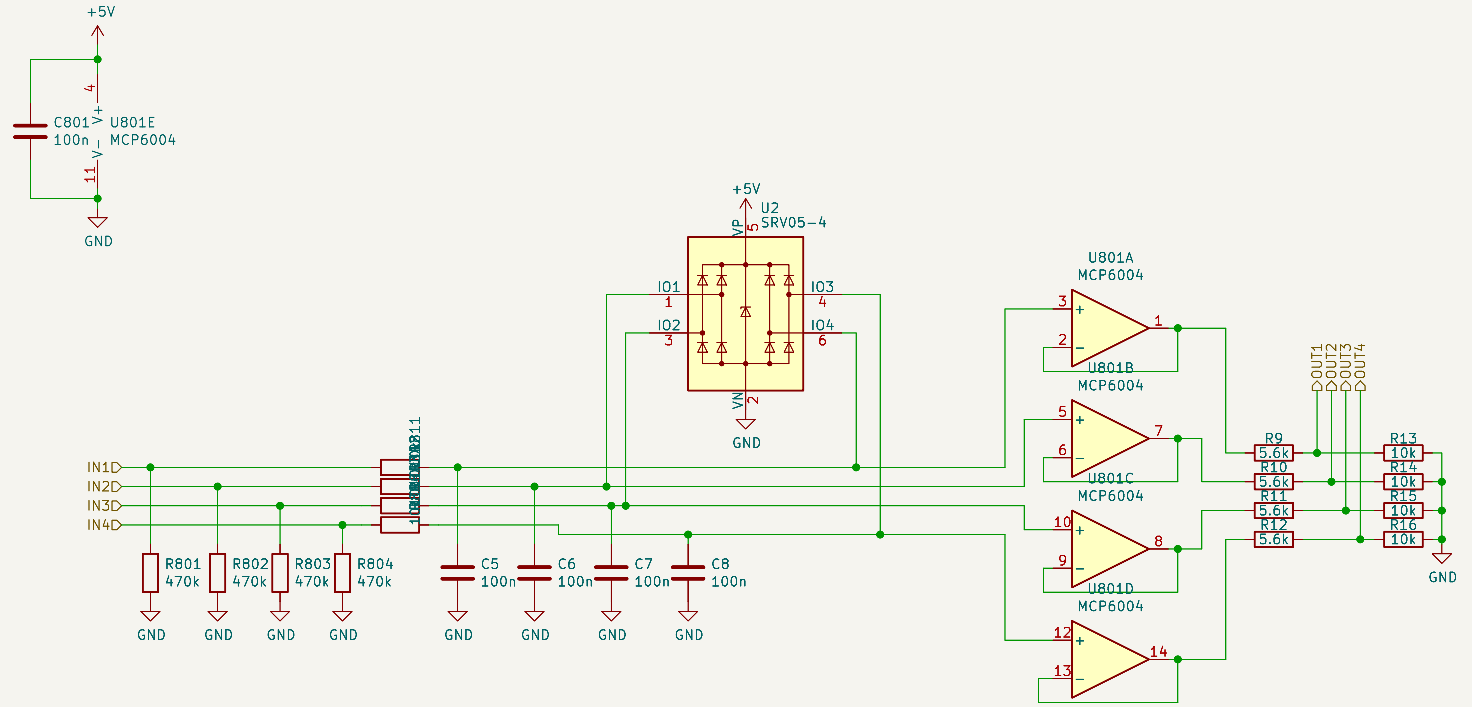
rusefi.com/s/proteus
github.com/mck1117/proteus
baseado em rusEFI Proteus v0.7

Sheet: /quad_analog1/
File: quad_analog.kicad_sch

Title: Proteus-BR — quad_analog

Size: A4 | Date: 2026-06-11 | **Rev: v1**

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 16/19



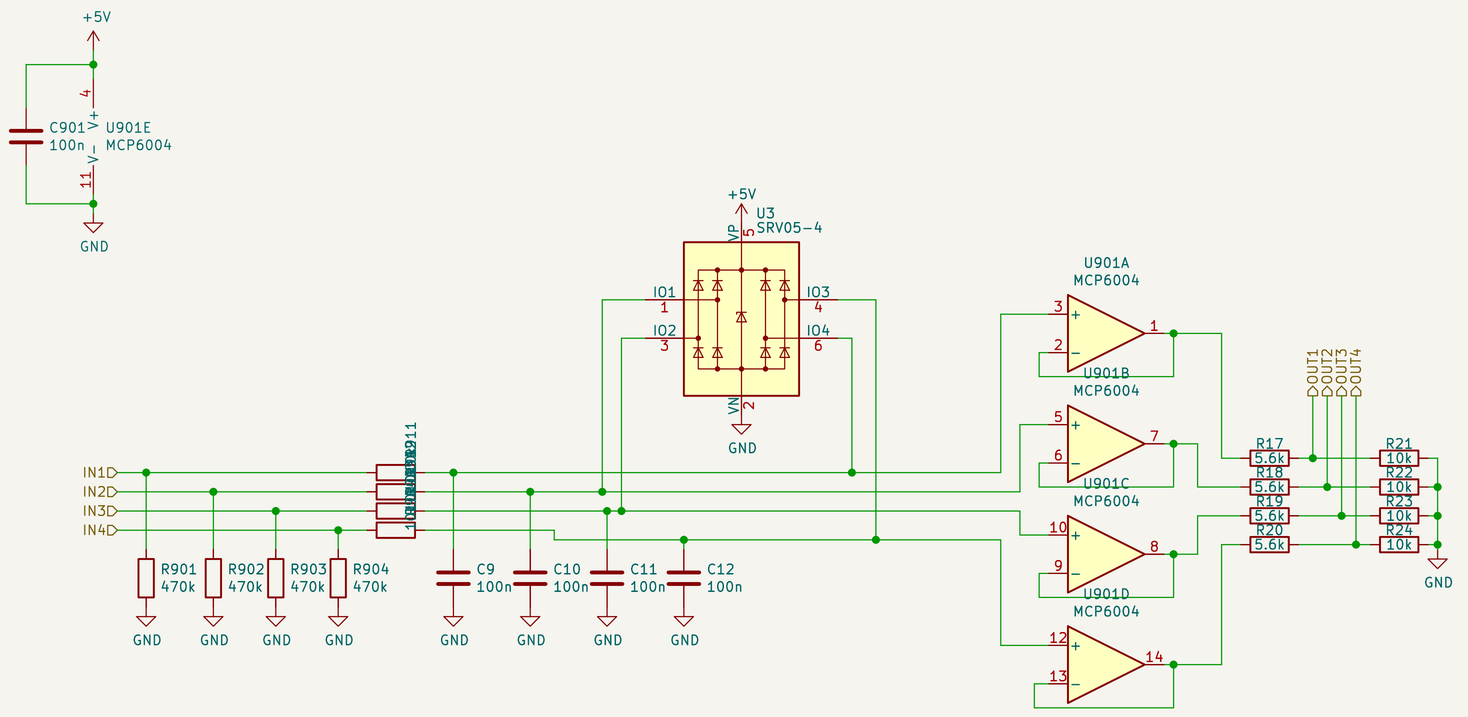
rusefi.com/s/proteus
github.com/mck1117/proteus
baseado em rusEFI Proteus v0.7

Sheet: /quad_analog2/
File: quad_analog.kicad_sch

Title: Proteus-BR — quad_analog

Size: A4 | Date: 2026-06-11 | **Rev: v1**

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 17/19

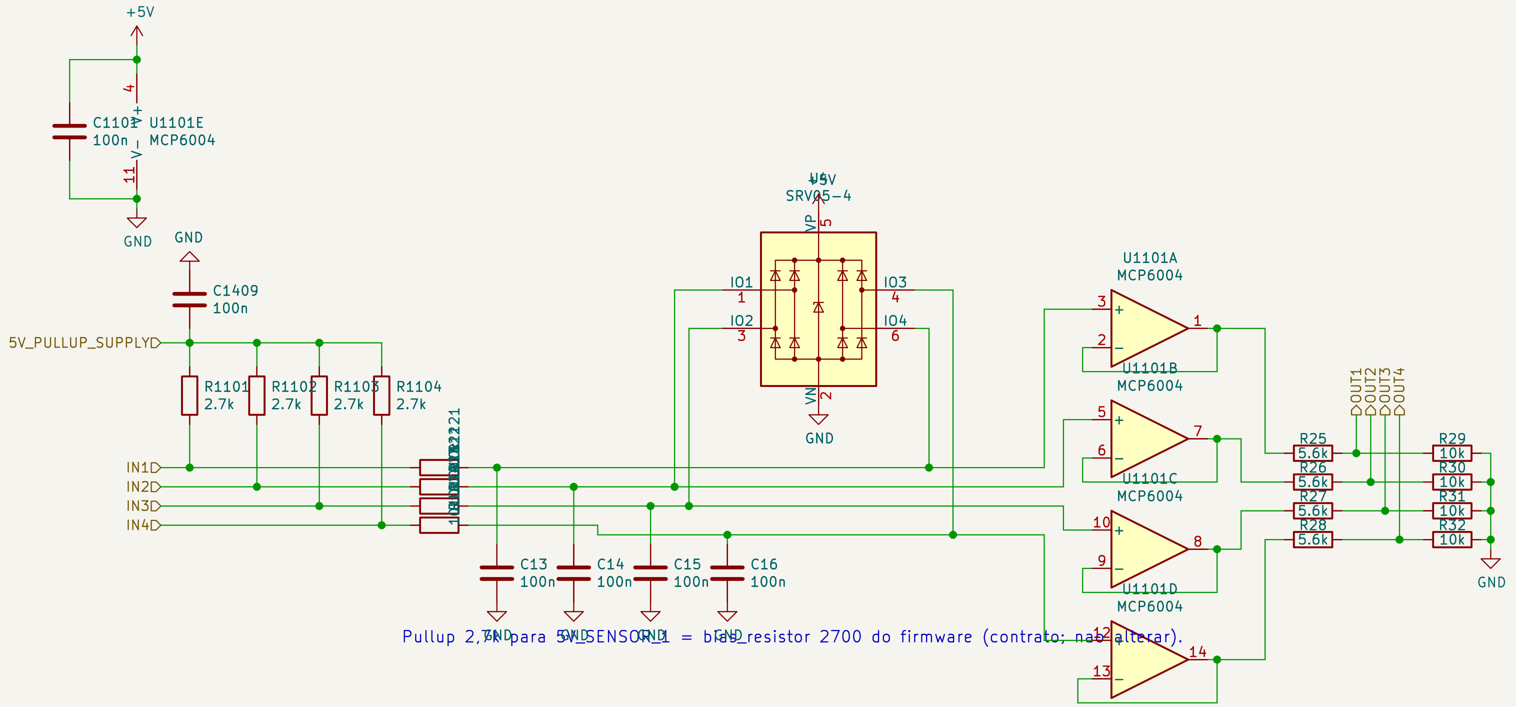


rusefi.com/s/proteus
github.com/mck1117/proteus
baseado em rusEFI Proteus v0.7

Sheet: /quad_analog3/
File: quad_analog.kicad_sch

Title: Proteus-BR — quad_analog

Size: A4	Date: 2026-06-11	**Rev: v1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 18/19



rusefi.com/s/proteus
github.com/mck1117/proteus
baseado em rusEFI Proteus v0.7

Sheet: /quad_analog_temp/
File: quad_analog_temp.kicad_sch

Title: Proteus-BR — quad_analog_temp

Size: A4	Date: 2026-06-11	**Rev: v1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 19/19